WRS for THEIA

WRS

Version 1.0, Draft 2

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Revision History

Name	Date	Reason for Changes	Version
Initial Draft	09/30/2026	Initial Draft	1.0, Draft 1
Final PP 1 Report	10/06/2026	Updated for our FPP1 submission.	1.0, Draft 2

1. Introduction

1.1. Purpose

This document establishes the formal requirements for Project THEIA. THEIA is a smartphone application designed to assist blind and visually impaired individuals with indoor navigation. This document transforms preliminary definitions into detailed system specifications. It also has traceability from existing navigational problems to specific system requirements.

1.2. Scope

The scope of Project THEIA includes developing requirements and a conceptual prototype for a smartphone application. The application will guide users from one indoor location to another across connected buildings. The scope covers functional requirements like routing and obstacle detection. It also covers non-functional requirements like safety and screen free usability. The project focuses on software requirements engineering and a graphical mock-up. It does not include building physical hardware or backend infrastructure.

1.3. Objectives and Success Criteria

The primary objective is to define a system that ensures safe and fast indoor navigation for visually impaired users. An important objective is designing an interface that relies on voice and audio feedback instead of screen readability. Success for this phase is achieved by delivering a document with traceability, connecting user problems to goals, objectives, and requirements. Providing a prototype and user manual are also key to project success.

1.4. Definitions, Acronyms, and Abbreviations

- AS-IS/TO-BE Scenarios: A method presenting system features by comparing the current state to the desired future state.
- Gen AI: Generative Artificial Intelligence tools used to assist the development team.
- GUI: Graphical User Interface representing the visual and interactive mock-up.
- NFRs: Non-Functional Requirements detailing system goals like speed and safety.
- THEIA: The indoor navigation smartphone application being developed.
- WRS: World Requirement Specification document serving as the main project deliverable.

1.5. Overview

The remainder of this document details the requirement engineering process for the THEIA application. It identifies issues within the preliminary project definitions and proposes design resolutions. It outlines an improved understanding of the system through a structured breakdown of problems, goals, objectives, and requirements. The document concludes by detailing the system prototype and how users interact with the application.

2. Preliminary Definition

2.1. Preliminary Domain

PD_ID	Preliminary Domain Description
PD-1	The system operates inside complex multi-level buildings like university halls and transit hubs.
PD-2	The primary users are individuals with severe visual impairments navigating unfamiliar indoor spaces.
PD-3	Secondary users include family members who manage app settings and security officers who get distress signals.

2.2. Preliminary Functional Requirements

PFR_ID	Preliminary FR Description
PFR-1	The app accepts destination inputs through voice commands and predicts frequent trips based on user schedules.
PFR-2	The system finds optimal indoor paths and offers multiple route choices to the user.
PFR-3	The app has real-time audio instructions indicating step counts and where to turn.
PFR-4	The system uses phone sensors to scan for physical barriers and gives avoidance instructions.
PFR-5	The app automatically contacts designated helpers or emergency services if a sudden fall occurs.

2.3. Preliminary Non-Functional Requirements

PNFR_ID	Preliminary NFR Description
PNFR-1	The app prioritizes user physical safety by selecting routes with the fewest obstacles.
PNFR-2	The system has a screen free experience, using voice and tactile feedback.
PNFR-3	The app allows users to edit audio preferences like voice volume and instruction frequency.
PNFR-4	The system architecture supports seamless updates for new languages and advanced hardware sensors.
PNFR-5	The app has fast response times and high reliability even in areas with poor network connectivity.

3. Issues with the Preliminary Definition Given

3.1. Domain Issues

Domain Issue ID	Domain Issue Description	
DI-1	PD_ID	PD-1
	The definition assumes the user is already inside the building and doesn't account for the transition from outside to inside.	
	Option 1	Limit the scope to indoor spaces, starting from the main lobby.
	Option 2	Include building entrances and nearby exterior pathways in the domain.
	Choice	Option 2
	Rationale	Users must successfully locate and enter the building to begin indoor navigation.
	Satisfied By	P-1
DI-2	PD_ID	PD-3
	Relying on security officers assume destination buildings have active security monitoring teams.	
	Option 1	Require buildings to have security systems for the app to function fully.
	Option 2	Increase distress signal list to include local emergency dispatch services.
	Choice	Option 2
	Rationale	Not all buildings have security staff available to assist the visually impaired during an emergency.
	Satisfied By	P-2
DI-3	PD_ID	PD-1
	The domain description assumes buildings and indoor paths are permanently open and ignores temporary closures for construction or maintenance.	
	Option 1	Integrate with the facility management systems to pull live building closure data.
	Option 2	Require users to report closed paths manually.
	Choice	Option 1
	Rationale	Blind users can't safely verify if a pathway is under construction until they encounter a physical barricade.
	Satisfied By	P-3
DI-4	PD_ID	PD-1
	The domain includes navigating to bathrooms but lacks detail to distinguish between specific facilities like gender-specific restrooms.	
	Option 1	Route users to the nearest bathroom regardless of type.
	Option 2	Include facility attributes in the domain map to allow specific routing.

	Choice	Option 2
	Rationale	Users require specific facilities and routing them to the wrong one causes frustration.
	Satisfied By	P-4

3.2. Functional Requirements Issues

FR Issue ID	Description	
FRI-1	PFR_ID	PFR-1
	Voice commands fail to be detected in noisy environments like crowded university hallways or busy transit hubs.	
	Option 1	Provide a secondary tactile input method for selecting frequent destinations.
	Option 2	Require the user to locate a quiet area to input navigation commands.
	Choice	Option 1
	Rationale	Users need a reliable method to set a destination when voice recognition fails due to background noise.
	Satisfied By	P-5
FRI-2	PFR_ID	PFR-4
	Common smartphone sensors cannot detect overhead obstacles like low hanging signs or open windows.	
	Option 1	Include an audio warning stating that the app only detects floor level physical obstacles.
	Option 2	Require the user to angle the phone upwards every so often to scan for overhead hazards.
	Choice	Option 1
	Rationale	Forcing users to move their phone upwards alters their walking posture and can compromise their balance.
	Satisfied By	P-6
FRI-3	PFR_ID	PFR-3
	Providing distance using time or step counts is inaccurate because walking speeds and length of stride vary among users.	
	Option 1	Calibrate step length during initial setup.
	Option 2	Provide distance in standard metric or imperial units like meters or feet.
	Choice	Option 2
	Rationale	Standard measurements offer precise distances that users can understand better than arbitrary step counts.
	Satisfied By	P-7
FRI-4	PFR_ID	PFR-5
	Automatically calling emergency services when the system loses location tracking results in frequent false alarms in areas with poor signal.	
	Option 1	Remove location loss as a trigger for automatic emergency calls.

	Option 2	Prompt the user to confirm they need help before placing the call.
	Choice	Option 2
	Rationale	Prompting the user prevents false alarms while still offering a reliable way to request help if they are lost.
	Satisfied By	P-8

3.3. Non-Functional Requirements (NFR) Issues

NFR Issue ID	Description	
NFRI-1	PNFR_ID	PNFR-2
	A screen free experience ignores users who have low vision and rely on visual aids for navigation.	
	Option 1	Keep the app screen free to focus development entirely on audio routing features.
	Option 2	Include high contrast visual elements and large text on the screen to assist low vision users.
	Choice	Option 2
	Rationale	Many legally blind individuals have some vision and benefit from bold visual cues with audio feedback.
	Satisfied By	P-9
NFRI-2	PNFR_ID	PNFR-5
	The preliminary requirement fails to define metrics for what defines a fast response time.	
	Option 1	Define fast response time as under a second for critical navigation alerts and obstacle warnings.
	Option 2	Define fast response time as under five seconds for all application interactions.
	Choice	Option 1
	Rationale	Delayed obstacle warnings compromise user physical safety while they move through a space.
	Satisfied By	P-10
NFRI-3	PNFR_ID	PNFR-4
	The preliminary requirement for the system to be widespread is too vague and nearly impossible to verify.	
	Option 1	Remove the requirement entirely.
	Option 2	Redefine "widespread” to mean having compatibility with major smartphone OSs like iOS and Android.
	Choice	Option 2
	Rationale	Specifying OS compatibility provides a measurable goal for development.
	Satisfied By	P-11
NFRI-4	PNFR_ID	PNFR-4
	The requirement for a “comfortable” route is subjective and can’t be programmed without specific parameters being provided.	

	Option 1	Define “comfortable” as routes with elevators instead of stairs.
	Option 2	Define “comfortable” as routes that are the shortest distance.
	Choice	Option 1
	Rationale	Many users with visual impairments prefer avoiding stairs to reduce the risk of tripping/falling.
	Satisfied By	P-12

4. WRS

4.1. W

4.2. Problem

Problem ID	Problem Description	Corresponding Goals
P-1	Users struggle with transitioning from outdoor spaces to indoor navigation routes.	G-1
P-2	Users lack reliable emergency support in buildings without security staff.	G-2
P-3	Users encounter unexpected pathway closures that regular building maps don’t include.	G-3
P-4	Users cannot navigate to specific facility types due to generalized mapping data.	G-4
P-5	Voice commands fail to register in loud environments.	G-5
P-6	Users are vulnerable to overhead obstacles because standard sensors only scan floor level.	G-6
P-7	Navigation instructions using step counts confuse users due to ambiguous stride lengths.	G-7
P-8	Automatic emergency calls trigger false alarms when location tracking drops temporarily.	G-8
P-9	Screen free interfaces exclude low vision users who need high contrast visual cues.	G-9
P-10	Delayed system response times place moving users in physical danger.	G-10
P-11	Vague requirements prevent targeted development for major smartphone OSs.	G-11
P-12	Routing algorithms fail to provide accessible paths because “comfort” doesn't have a clear technical definition.	G-12

4.3. Goals

Goal ID	Goal Description	Backward Traceability	Forward Traceability
G-1	The system shall include building entrances and exterior pathways in the navigation domain.	P-1	IFRO-1
G-2	The system shall connect to local emergency dispatch services for distress signals.	P-2	IFRO-2
G-3	The system shall integrate with facility management databases to identify temporary closures.	P-3	IFRO-3
G-4	The system shall include specific facility attributes in the mapping data.	P-4	IFRO-4
G-5	The system shall provide tactile input methods alongside voice recognition.	P-5	IFRO-5, IFRO-10, INFRO-5

G-6	The system shall warn users that it only detects floor level obstacles.	P-6	IFRO-6
G-7	The system shall provide distance instructions using standard measurement units.	P-7	IFRO-7
G-8	The system shall prompt users for confirmation before placing emergency calls.	P-8	IFRO-8
G-9	The system shall include a high contrast visual interface alongside audio routing.	P-9	IFRO-10, INFRO-1, INFRO-5
G-10	The system shall process critical navigation alerts in under one second.	P-10	INFRO-2
G-11	The system shall be compatible with iOS and Android OSs.	P-11	INFRO-3
G-12	The system shall define comfortable routes as paths that avoid stairs.	P-12	IFRO-9, INFRO-4

4.4. Improved Understanding of Domain, Stakeholders, Functional, and Non-Functional Objectives

4.5. Improved Domain

Improved Domain ID	Improved Domain Description
ID-1	The operating domain encompasses indoor building layouts across multilevel campus facilities and transit hubs, including immediate exterior pathways and primary accessible entrance doors.
ID-2	The domain includes architectural features like rooms, hallways, elevators, stairs, and ramps, as well as dynamic conditions including temporary construction closures and maintenance barricades.
ID-3	Navigational points of interest within the domain incorporate finer facility attributes to distinguish specific amenities like accessible restrooms and administrative service desks.

4.6. Stakeholders

- Primary Stakeholders
  - Blind and visually impaired individuals who need to navigate indoor spaces.
- Secondary Stakeholders
  - Facility management personnel who provide building mapping data and report temporary pathway closures.
  - Caretakers and family members who configure the application and provide emergency assistance.
  - Police and local emergency dispatch services who get distress signals.
  - Accessibility department, which oversees building accessibility and supports users.

4.7. Improved Functional Objectives

Improved FR Objective ID	Objective Description	Alleviates Problems	Achieves Goals
IFRO-1	Guide the user along exterior pathways to the building entrance and continue into indoor navigation without a break.	P-1	G-1
IFRO-2	Notify designated helpers and local emergency dispatch of a distress event, with the user's location.	P-2	G-2
IFRO-3	Use facility management for closure data to avoid closed paths and reroute the user when a closure appears.	P-3	G-3
IFRO-4	Let the user route to a specific facility type using facility attributes in the map.	P-4	G-4
IFRO-5	Let the user set a destination without relying only on voice, using tactile input and trip suggestions.	P-5	G-5
IFRO-6	Detect floor-level obstacles, give avoidance instructions, and tell the user that overhead hazards are not detected.	P-6	G-6
IFRO-7	Give distance and turn instructions in standards units.	P-7	G-7
IFRO-8	Confirm with the user before placing any emergency call, to prevent false alarms.	P-8	G-8
IFRO-9	Present route alternatives, including a comfortable option such as stair-free, and accept user's choice.	P-12	G-12
IFRO-10	Let users and their caretakers adjust audio and visual preferences and emergency contacts.	P-5, P-9	G-5, G-9

4.8. Improved Non-Functional Objectives

Improved NFR Objective ID	Objective Description	Alleviates Problems	Achieves Goals
INFRO-1	Support both audio/haptic-only use and low-vision use, with a high-contrast, large-text visual interface.	P-9	G-9
INFRO-2	Deliver critical alerts fast and reliably, processing them on the device so poor connectivity does not delay the.	P-10	G-10
INFRO-3	Be compatible with IOS and Android and be extensible with new languages and sensors.	P-11	G-11

INFRO-4	Define a comfortable route as one that avoids stairs and choose safe routes that minimize known obstacles.	P-12	G-12
INFRO-5	Let users personalize audio and visual output.	P-5, P-9	G-5, G-9

4.9. RS

4.10. Functional Requirements

FR ID	Description
FR-1	The system shall guide the user along exterior pathways from their location to the entrance of the destination building.
FR-2	The system shall announce when the user reaches the building entrance and continue into indoor guidance without further input.
FR-3	The system shall contact the user's designated helpers and local emergency dispatch, with the user's last known location. When the user requests help, a fall is detected, or the user is lost.
FR-4	The system shall ask the user to confirm before placing an emergency call.
FR-5	The system shall obtain temporary closure data from facility management systems.
FR-6	The system shall notify the user and reroute when a closure affects the active route.
FR-7	The system shall store facility attributes and let the user request a destination by facility type.
FR-8	The system shall accept destinations by voice command.
FR-9	The system shall accept destinations through tactile input from a list of saved and frequent destinations.
FR-10	The system shall suggest likely trips based on the user's schedule and history and start navigation only after the user confirms.
FR-11	The system shall compute routes and, when several exist, present the options and accept the user's choice.
FR-12	The system shall use phone sensors to detect floor-level obstacles and give avoidance instructions.
FR-13	The system shall tell the user that only floor-level obstacles are detected.
FR-14	The system shall state distances in meters or feet, per the user's setting.
FR-15	The system shall give a turn warning ahead of each turn and a stop-and-turn cue at the turn point.
FR-16	The system shall provide a high-contrast, large-text visual interface alongside audio and haptic feedback.
FR-17	The system shall let the user adjust text size and contrast.
FR-18	The system shall offer a comfortable route option that avoids stairs.

FR-19	The system shall let the user edit audio preferences, including voice volume, speech rate, and instruction frequency.
FR-20	The system shall let an authorized caretaker edit emergency contacts and settings, with the user's approval.

4.11. Non-Functional Requirements

NFR ID	Non-Functional Requirement
NFR-1	The visual interface shall use a contrast ratio of at least 7:1 and text scalable to 200% or more. Every core function shall also be usable with audio and haptic feedback alone.
Operationalized Functional Requirements	FR-9, FR-16, FR-17
Satisfies Non-Functional Requirement Issue	NFRI-1
Satisfies Non-Functional Objective	INFRO-1
Constraints	Must work with VoiceOver and TalkBack. Audio and haptic output must never depend on the visual channel.
Satisfied by Prototype Feature	PF-1, PF-4, PF-7
NFR ID	**Non-Functional Requirement**
NFR-2	Critical navigation alerts and obstacle warnings shall be presented within 1 second of detection.
Operationalized Functional Requirements	FR-6, FR-12, FR-15
Satisfies Non-Functional Requirement Issue	NFRI-2
Satisfies Non-Functional Objective	INFRO-2
Constraints	Critical alerts must be processed on the device. Sensor quality varies by phone model.
Satisfied by Prototype Feature	PF-4, PF-5
NFR ID	**Non-Functional Requirement**
NFR-3	Navigation, turn guidance, and obstacle detection shall keep working without network connectivity, using maps downloaded in advance.

Operationalized Functional Requirements	FR-5, FR-12, FR-15
Satisfies Non-Functional Requirement Issue	NFRI-2
Satisfies Non-Functional Objective	INFRO-2
Constraints	Maps must be downloaded before use. Closure data may be out of date offline, and the user shall be told so.
Satisfied by Prototype Feature	PF-9, PF-12
NFR ID	**Non-Functional Requirement**
NFR-4	The app shall run on current and previous major versions of iOS and Android with equivalent core functionality.
Operationalized Functional Requirements	FR-1 to FR-20
Satisfies Non-Functional Requirement Issue	NFRI-3
Satisfies Non-Functional Objective	INFRO-3
Constraints	Must follow Apple and Google platform and accessibility guidelines. Available sensors differ by platform.
Satisfied by Prototype Feature	PF-1 to PF-13
NFR ID	**Non-Functional Requirement**
NFR-5	Comfortable routes shall contain no stairs, using elevators or ramps. If no stair-free route exists, the system shall say so and ask the user. Route planning shall also avoid known obstacles and closures.
Operationalized Functional Requirements	FR-6, FR-11, FR-18
Satisfies Non-Functional Requirement Issue	NFRI-4
Satisfies Non-Functional Objective	INFRO-4

Constraints	The map must include connector attributes (stairs, elevator, ramp). Stair-free routes may take longer.
Satisfied by Prototype Feature	PF-3, PF-7
NFR ID	**Non-Functional Requirement**
NFR-6	Audio and visual preference changes shall apply immediately and persist across sessions. Volume shall be adjustable from 0 to 100% and instruction frequency shall have at least three levels.
Operationalized Functional Requirements	FR-14, FR-17, FR-19
Satisfies Non-Functional Requirement Issue	NFRI-3
Satisfies Non-Functional Objective	INFRO-5
Constraints	Hardware volume output limits apply. Changing settings very quickly shouldn't crash or block audio commands.
Satisfied by Prototype Feature	PF-7
NFR ID	**Non-Functional Requirement**
NFR-7	New languages and sensors shall be addable through defined extension points without changing the core routing and guidance logic.
Operationalized Functional Requirements	FR-8, FR-19
Satisfies Non-Functional Requirement Issue	NFRI-3
Satisfies Non-Functional Objective	INFRO-3
Constraints	Supporting adding different languages through language packs depends on the OS's support for text to speech engine availability.
Satisfied by Prototype Feature	PF-7

4.12. Specifications

Functional Specification ID	Functional Requirement

FS-1	The application guides the user along outdoor paths directly to the accessible front door of their destination building. The system monitors live coordinates and calls out clear walking directions, so the user never gets turned around before stepping inside.
Satisfies Functional Requirement	FR-1
Satisfies Objectives	IFRO-1
Satisfied by Prototype Feature	PF-1
Functional Specification ID	**Functional Requirement**
FS-2	When the user steps through the doorway, the app switches from outdoor guidance to indoor navigation on its own. The audio prompts keep playing smoothly, so the user does not have to stop walking or touch the screen to reload maps.
Satisfies Functional Requirement	FR-2
Satisfies Objectives	IFRO-1
Satisfied by Prototype Feature	PF-2
Functional Specification ID	**Functional Requirement**
FS-3	In an emergency, the app sends out an alert containing the user's current floor, hallway coordinates, and a timestamp. This message goes to designated family members and local dispatch, so responders know the exact indoor room location.
Satisfies Functional Requirement	FR-3
Satisfies Objectives	IFRO-2
Satisfied by Prototype Feature	PF-6
Functional Specification ID	**Functional Requirement**
FS-4	Before placing an emergency call, the app vibrates and asks the user out loud to confirm they need help. The user has a few seconds to cancel the alert with a single screen tap or spoken word, preventing false alarms if the phone simply slipped out of a hand.
Satisfies Functional Requirement	FR-4
Satisfies Objectives	IFRO-8
Satisfied by Prototype	PF-6

Feature	
Functional Specification ID	**Functional Requirement**
FS-5	Every time the application starts up, it checks in with the campus maintenance database to grab updates on hallway construction, repairs, or closed corridors. The app marks these blocked areas on the internal map right away.
Satisfies Functional Requirement	FR-5
Satisfies Objectives	IFRO-3
Satisfied by Prototype Feature	PF-8
Functional Specification ID	**Functional Requirement**
FS-6	If a route runs into an unexpected hallway closure, the app spots the blockage immediately. It works out a detour around the closed space within half a second and tells the user which way to turn to stay on a clear path.
Satisfies Functional Requirement	FR-6
Satisfies Objectives	IFRO-3
Satisfied by Prototype Feature	PF-3, PF-4
Functional Specification ID	**Functional Requirement**
FS-7	The system marks every indoor destination with clear tags, noting details like single-occupancy restrooms, accessible stalls, or specific student service desks. Users can request a specific type of room rather than ending up at a general doorway that does not fit their needs.
Satisfies Functional Requirement	FR-7
Satisfies Objectives	IFRO-4
Satisfied by Prototype Feature	PF-1, PF-7
Functional Specification ID	**Functional Requirement**
FS-8	The app listens through the phone microphone and uses voice processing right on the device to turn spoken words into destination searches. This lets users say where they want to go naturally without struggling to type on a flat glass screen.
Satisfies Functional Requirement	FR-8

Satisfies Objectives	IFRO-5
Satisfied by Prototype Feature	PF-1
Functional Specification ID	**Functional Requirement**
FS-9	For noisy spaces where speech recognition struggles, the app provides a simple list of frequent destinations. Users can move through these saved spots using large, high-contrast touch areas and tap twice to start navigation.
Satisfies Functional Requirement	FR-9
Satisfies Objectives	IFRO-5
Satisfied by Prototype Feature	PF-1, PF-7
Functional Specification ID	**Functional Requirement**
FS-10	The app checks the user schedule and current time of day to suggest the most likely next destination. The user hears the recommendation out loud and can confirm it with one tap rather than searching through lists.
Satisfies Functional Requirement	FR-10
Satisfies Objectives	IFRO-5
Satisfied by Prototype Feature	PF-1
Functional Specification ID	**Functional Requirement**
FS-11	When multiple routes lead to the same room, the app presents up to three choices and reads out the total walking distance along with any stair counts. The user can pick their preferred route using voice or touch.
Satisfies Functional Requirement	FR-11
Satisfies Objectives	IFRO-9
Satisfied by Prototype Feature	PF-3
Functional Specification ID	**Functional Requirement**
FS-12	The camera scans the floor ahead while the user walks, watching for tripping hazards, dropped bags, or low barriers within two meters. If

	something sits in the path, the app calls out an immediate warning telling the user to step left or right.
Satisfies Functional Requirement	FR-12
Satisfies Objectives	IFRO-6
Satisfied by Prototype Feature	PF-4, PF-5
Functional Specification ID	**Functional Requirement**
FS-13	When a trip begins, the app speaks a clear reminder that its cameras and sensors only spot obstacles down on the floor. This prompt ensures users remember to keep using their white cane for upper body and head-level hazards.
Satisfies Functional Requirement	FR-13
Satisfies Objectives	IFRO-6
Satisfied by Prototype Feature	PF-1
Functional Specification ID	**Functional Requirement**
FS-14	The app speaks remaining distances in standard units like feet or meters depending on what the user selects in settings. This avoids the confusion of arbitrary step counts that change based on individual stride length.
Satisfies Functional Requirement	FR-14
Satisfies Objectives	IFRO-7
Satisfied by Prototype Feature	PF-4
Functional Specification ID	**Functional Requirement**
FS-15	The app gives users plenty of warning before a turn by announcing the corner two meters in advance. Right at the turn threshold, the phone provides a distinct vibration pulse alongside a spoken turn cue so the user turns with confidence.
Satisfies Functional Requirement	FR-15
Satisfies Objectives	IFRO-7
Satisfied by Prototype Feature	PF-4, PF-5

Functional Specification ID	Functional Requirement
FS-16	For users who still have partial sight, the app displays directional markers using bold yellow shapes on a deep black background. The text scales up cleanly so low-vision users can follow visual arrows alongside audio cues.
Satisfies Functional Requirement	FR-16
Satisfies Objectives	INFRO-1
Satisfied by Prototype Feature	PF-1, PF-4, PF-7
Functional Specification ID	**Functional Requirement**
FS-17	The app saves all custom text sizes, color contrast choices, and display modes right to the phone storage. These preferences stay active every time the user opens the application.
Satisfies Functional Requirement	FR-17
Satisfies Objectives	INFRO-1, INFRO-5
Satisfied by Prototype Feature	PF-7
Functional Specification ID	**Functional Requirement**
FS-18	When a user selects a comfortable route, the navigation engine creates a path that completely skips staircases. The app routes users through elevators and ramps, warning them beforehand if a stair-free path is not available.
Satisfies Functional Requirement	FR-18
Satisfies Objectives	IFRO-9, INFRO-4
Satisfied by Prototype Feature	PF-3
Functional Specification ID	**Functional Requirement**
FS-19	Users can tailor voice feedback to fit their hearing preferences, adjusting volume, speech speed, and how often they want distance callouts. This keeps the guidance clear in loud hallways without becoming overwhelming.
Satisfies Functional Requirement	FR-19
Satisfies Objectives	INFRO-5

Satisfied by Prototype Feature	PF-7
Functional Specification ID	**Functional Requirement**
FS-20	Caretakers can configure emergency phone numbers and medical details inside a protected profile menu secured by a simple PIN. This keeps critical safety contacts accurate while keeping daily navigation controls clutter free.
Satisfies Functional Requirement	FR-20
Satisfies Objectives	IFRO-10
Satisfied by Prototype Feature	PF-7

5. Preliminary Prototype

5.1. Prototyped Features

Prototype Feature ID	Feature Name	Description
PF-1	Destination Selection Screen	Full screen voice recognition input display paired with high contrast, oversized buttons for frequent destinations.
PF-2	Entrance Transition Indicator	Audio confirmation banner and haptic feedback prompt signaling a shift from exterior pathways to interior floor maps.
PF-3	Route Selection Card	Audio and visual interface presenting up to three calculated route options with elevator indicators and distance summaries.
PF-4	Turn by Turn Guidance Display	Navigation screen, displaying large direction arrows, distance readouts, and active spoken walking instructions.
PF-5	Obstacle Warning Banner	High contrast alert overlay accompanied by distinct emergency vibration patterns for detected floor hazards.
PF-6	Emergency Alert Modal	Automated ten second countdown screen with one tap cancellation and voice abort controls before placing emergency calls.
PF-7	Preferences and Caretaker Panel	Protected settings screen featuring sliders for volume, speech rate, font scaling, contrast modes, and caretaker PIN entry.
PF-8	Maintenance Reroute Notification	Visual banner and voice prompt alerting the user to dynamic path rerouting caused by campus construction closures.
PF-9	Offline Map Cache Manager	Background data manager that stores building geometries locally to support dead reckoning without active cellular data.
PF-10	Voice Command Query Processor	Audio interface feedback ring showing system listening states during speech input.
PF-11	Step Calibration Utility	Calibration screen (optional) for users who prefer pairing standard distance cues with individual walking cadences.
PF-12	Signal Reacquisition Screen	High-priority interface that speaks a stop instruction if internal tracking confidence drops below acceptable levels.
PF-13	Emergency Contact Setup Screen	Secondary profile interface allowing family members to define dispatch recipients and trusted helper phone numbers.

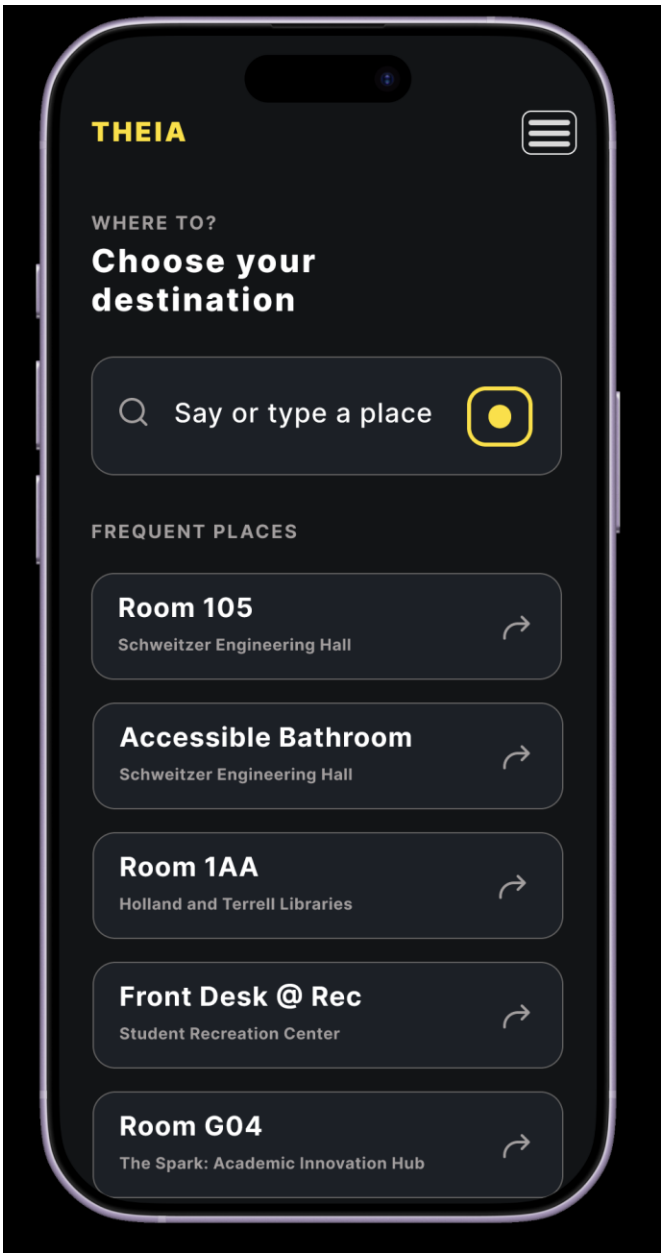


Figure 5.1 (PF-1, FR-7, FR-8, FR-9, NFR-1)

Destination Selection Screen. This interface allows users to initiate a voice search or select from a high contrast tactile list of frequent indoor destinations, including specific rooms and facility types.

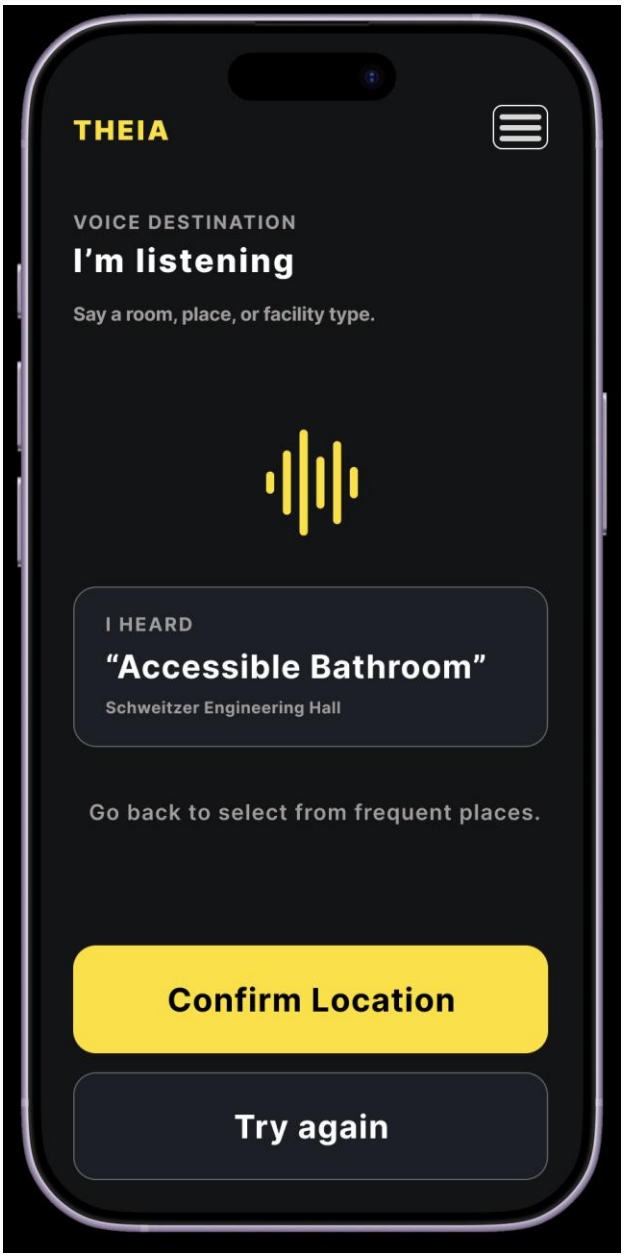


Figure 5.2 (PF-10, FR-8)

Voice Command Query Processor. This screen displays an active listening state for voice input, transcribes the spoken facility request, and prompts the user to visually or vocally confirm the destination before routing begins.



Figure 5.3 (PF-2, FR-2)

Entrance Transition Indicator. This alert notifies the user that they have successfully reached the exterior building entrance and confirms the automatic transition to indoor navigation guidance.

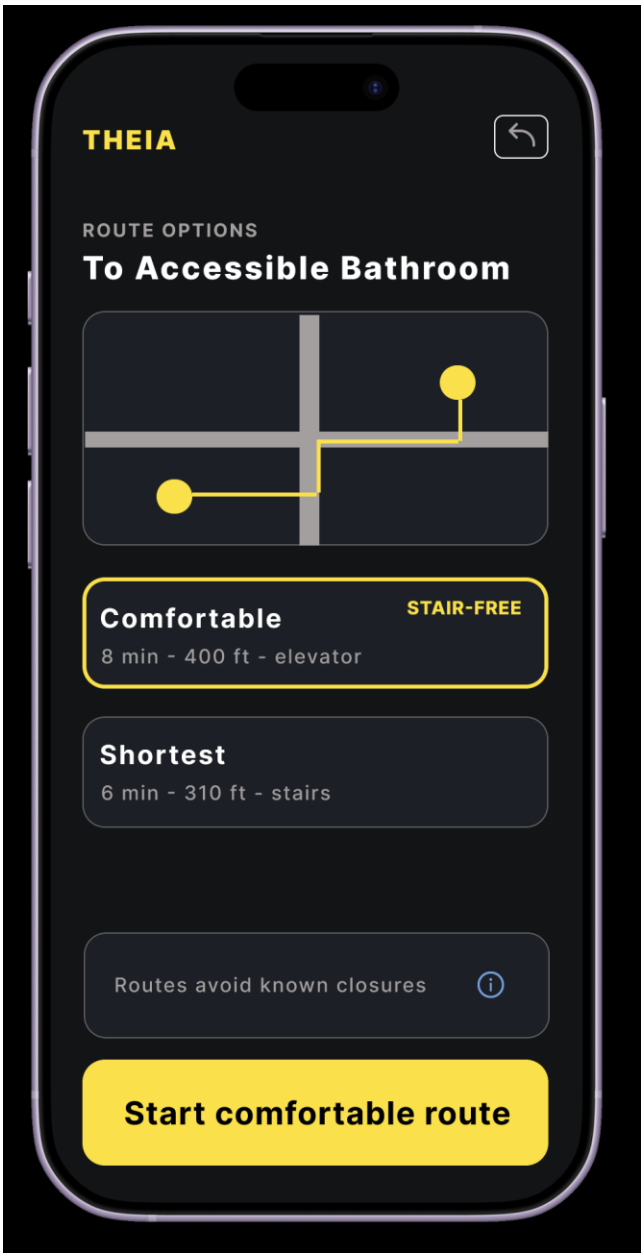


Figure 5.4 (PF-3, FR-6, FR-11, FR-18, NFR-5)

Route Selection Card. This interface presents multiple calculated indoor paths, highlighting a comfortable stair free option with elevators and confirming that all routes actively bypass known hallway closures.



Figure 5.5 (PF-4, FR-14, FR-15, NFR-1)

Turn by Turn Guidance Display. This screen shows the active indoor route with a large directional arrow and current distance readouts in feet. The interface previews the next upcoming turn and offers a high contrast button to repeat the latest audio instruction.



Figure 5.6 (PF-5, FR-12, FR-13, NFR-2)

Obstacle Warning Banner. The application pauses active navigation to display a red alert regarding a floor level hazard detected three feet ahead. The screen includes a reminder that overhead obstacles are not detected alongside options to call for help or repeat the warning.

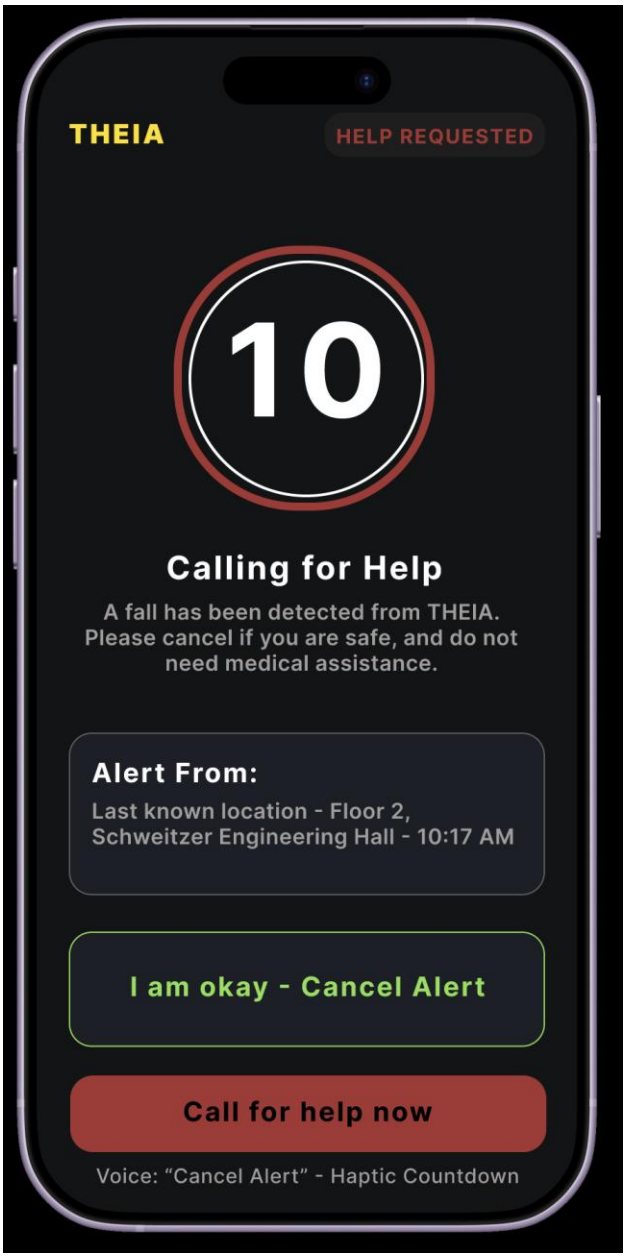


Figure 5.7 (PF-6, FR-3, FR-4)

Emergency Alert Modal. This interface triggers after detecting a potential fall and initiates a ten second haptic countdown. The user can cancel the automated emergency dispatch using a voice command or by tapping the designated cancellation button.

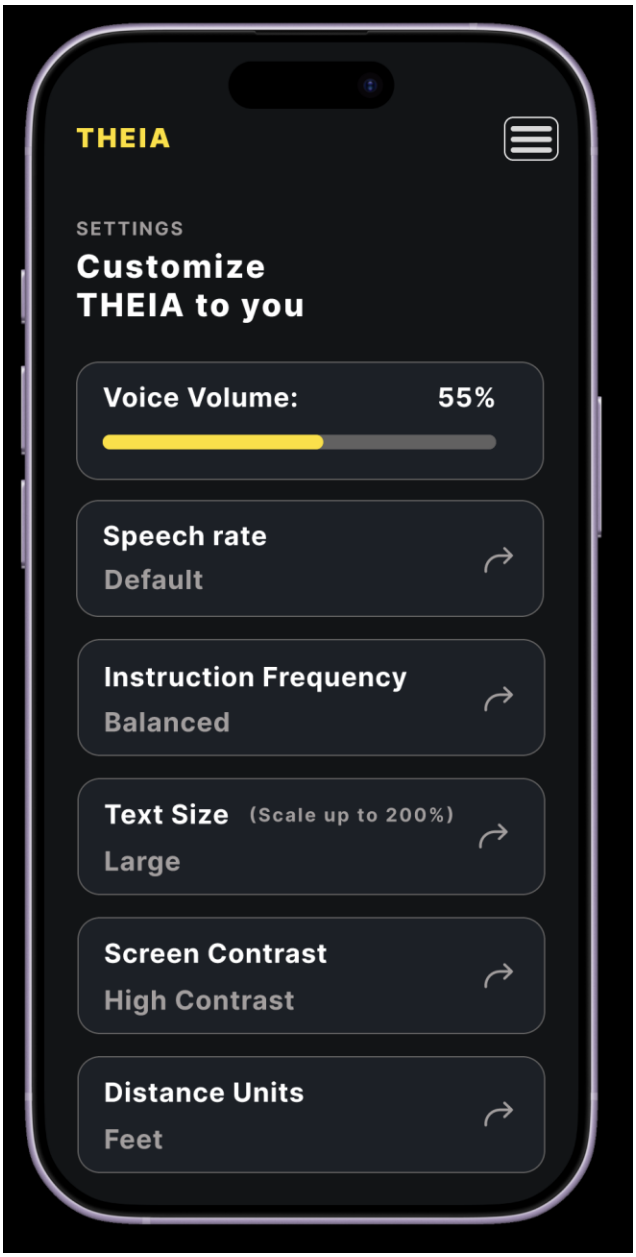


Figure 5.8 (PF-7, FR-14, FR-17, FR-19, NFR-6)

Preferences Panel. This settings menu allows the user to customize the application experience by adjusting voice volume, speech rate, instruction frequency, and text size. The user can also toggle high contrast visual modes and set their preferred distance units.

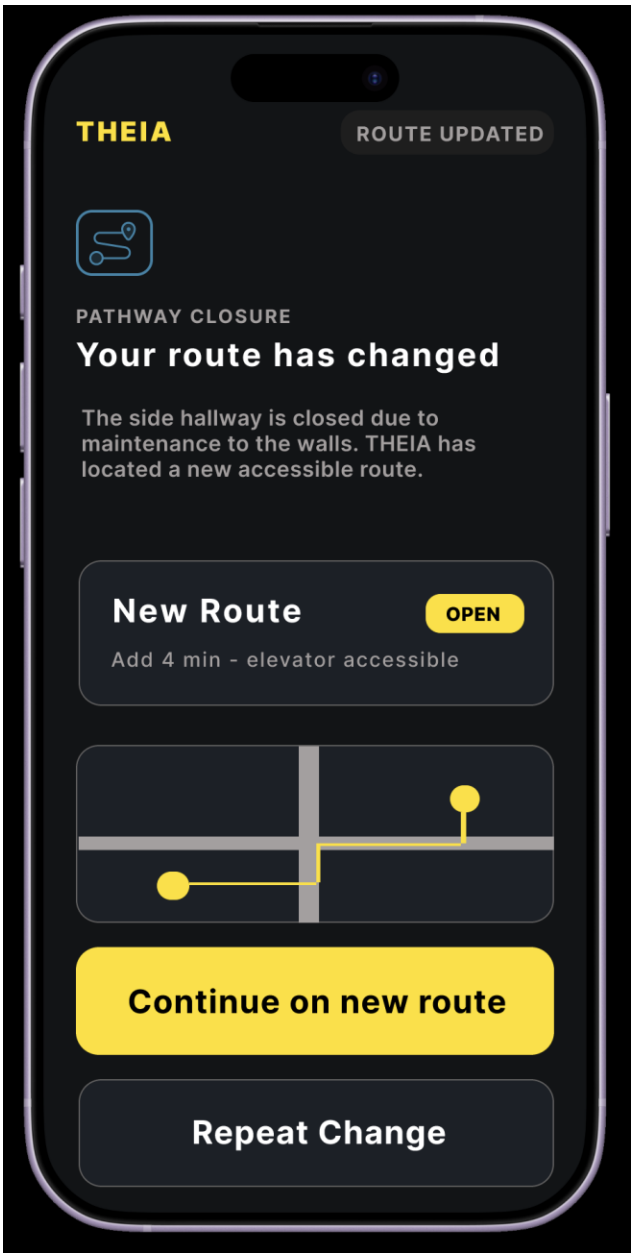


Figure 5.9 (PF-8, FR-5, FR-6)

Maintenance Reroute Notification. This screen alerts the user to an unexpected pathway closure caused by hallway maintenance. The interface immediately provides a new accessible detour path and details the added travel time.

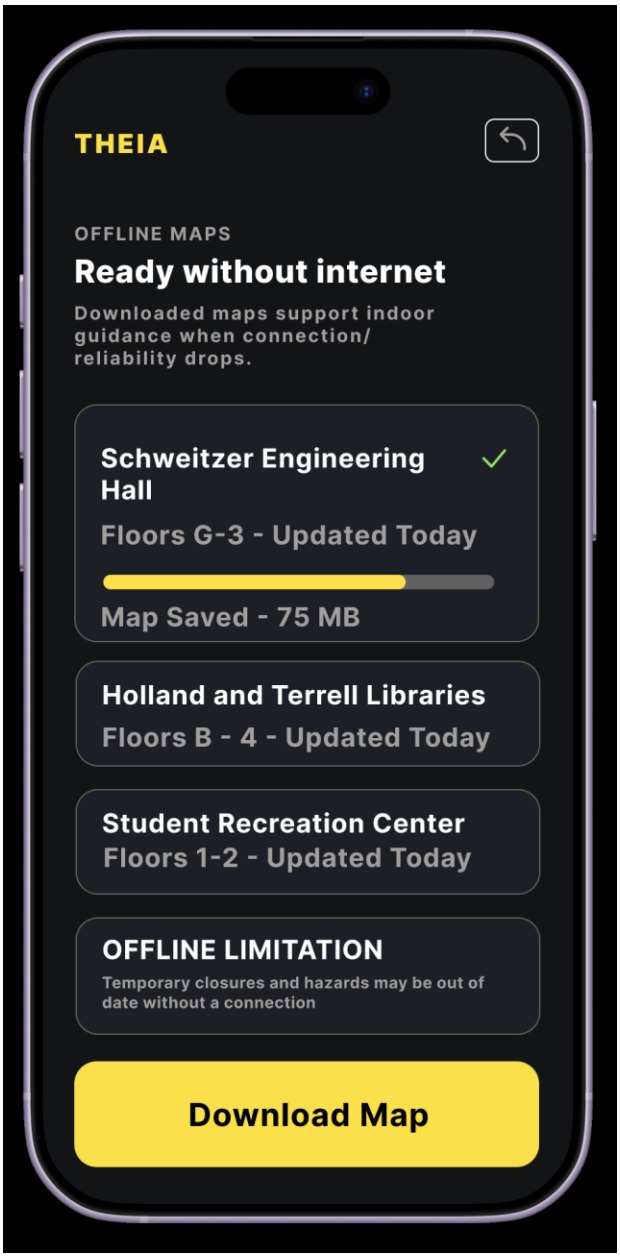


Figure 5.10 (PF-9, FR-5, NFR-3)

Offline Map Cache Manager. This screen allows users to download building floor plans for offline navigation. It also displays a clear warning that maintenance closures may be out of date when using the app without an active network connection.

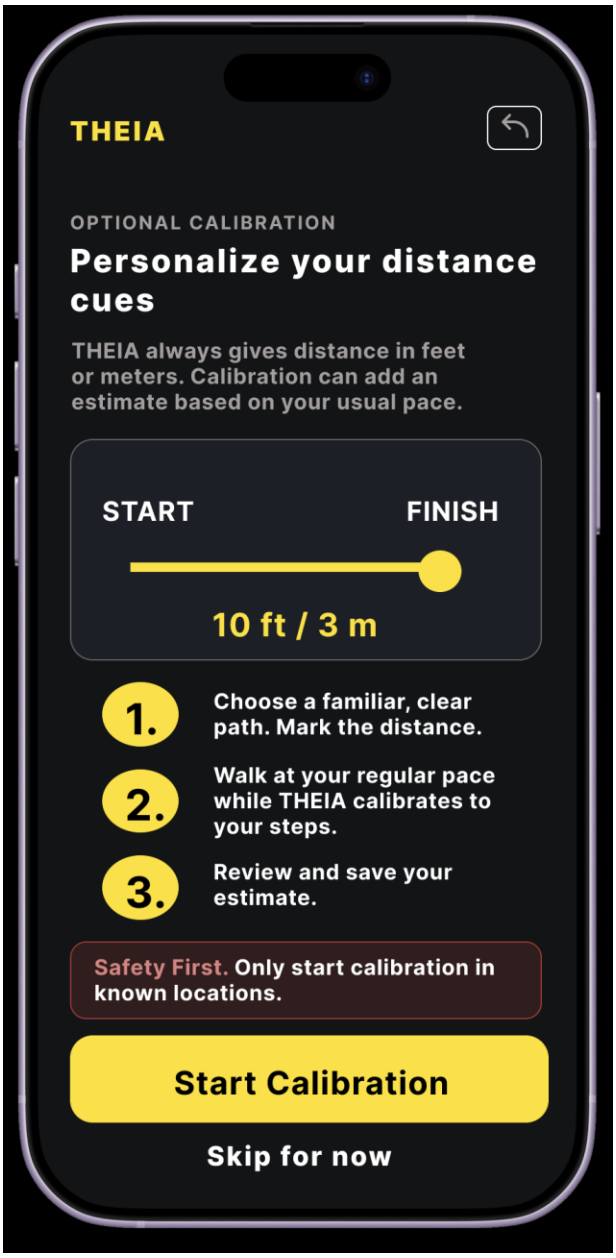


Figure 5.11 (PF-11, FR-14)

Optional Step Calibration. This interface allows users to walk a known distance to calibrate their personal step length. This optional feature can replace standard distance units with personalized step counts for users who prefer that metric.

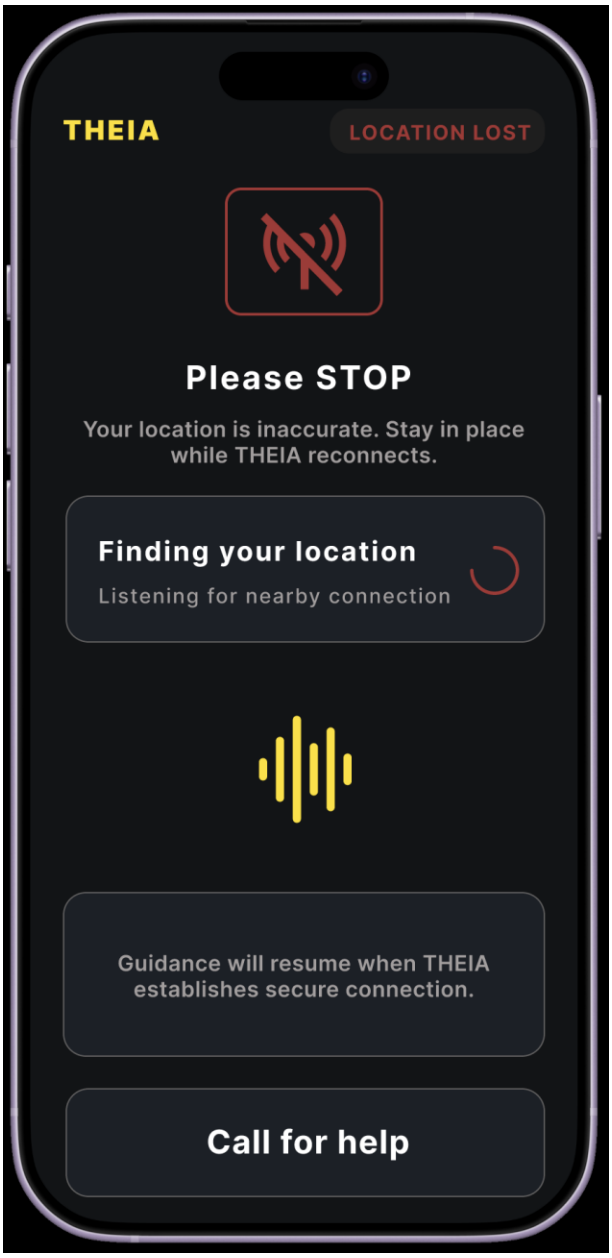


Figure 5.12 (PF-12, FR-3, NFR-3)

Signal Reacquisition Screen. This high-priority alert directs the user to stop walking immediately when indoor tracking confidence drops. It provides a quick option to call for help while the system stays in place and attempts to reconnect.

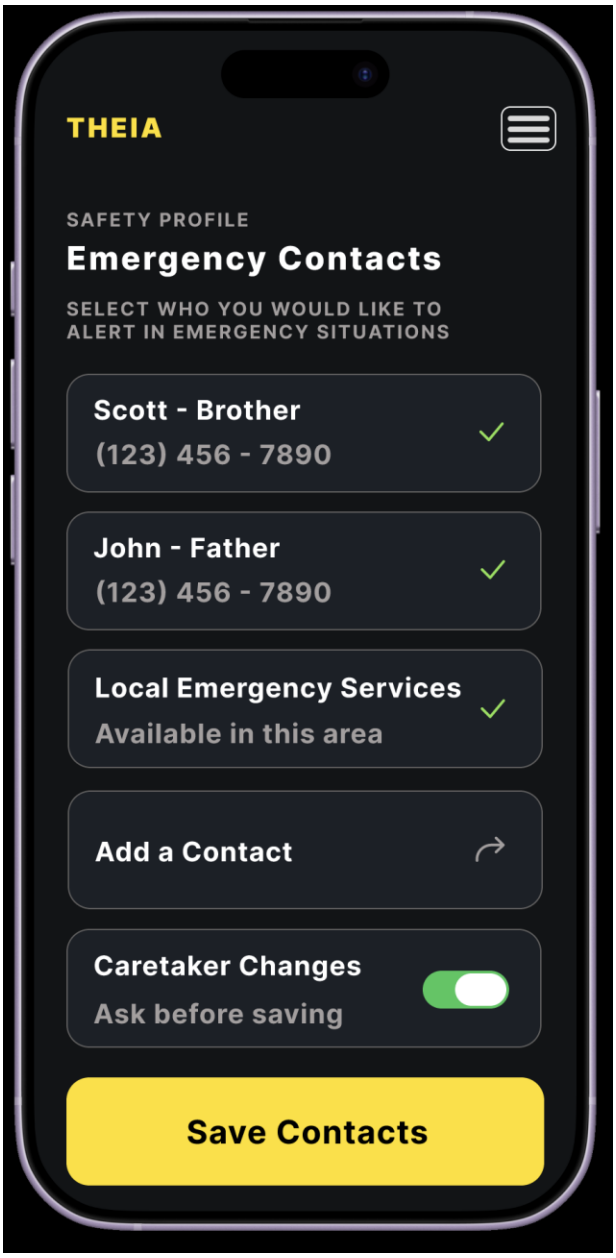


Figure 5.13 (PF-13, FR-3, FR-20)

Emergency Contact Setup. This profile menu allows users to select trusted family members and local emergency services to receive alerts during a distress event. It also includes a toggle requiring user approval for any changes made by a caretaker.

5.2. Scenario Walkthrough Overview

The preliminary prototype supports a primary scenario: navigating to an accessible bathroom inside Schweitzer Engineering Hall. The user journey spans setting preferences, selecting the destination, choosing a route, following turn-by-turn guidance, and handling dynamic alerts like floor hazards. Because the interface is designed around audio and haptic feedback, the complete

step-by-step instructions for operating these features and handling system alerts are in the external THEIA User Guide document.

6. References

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